SiPM charge response and provisional ASIC readout for NAUSICAA electroluminescence

NAUSICAA model and implementation note

4 November 2026

Abstract

The response begins with incident VUV photon counts at each SiPM in 250 ns buckets. A stochastic microcell model samples photodetection and dark pulses, follows the charge recovery of each microcell, and saves the sensor charge before applying electronics. The provisional ASIC reads bucket-average current, adds independent Gaussian noise of 0.5 photoelectron equivalents per bucket, and clips the output above 1 mA. This note states the adopted inputs, physical approximations and output contract, and derives the relation to AXEL's subsequent correction for SiPM nonlinearity. Numerical validation and a small pilot precede a full-sample response calculation.

1 Input, terminology and provenance

The input is the number $N_{\gamma,cb}$ of photons incident on SiPM channel c in bucket b of width Δt. The existing four HDF5 inputs represent `bb0nu` and `single_electrons`, each with pure-Xe and Xe–15% He transport and electroluminescence models. The gas files are alternative realizations of the detector response for the same underlying event sample. The current production uses $\Delta t = 0.25\ \mu\text{s} = 250$ ns, as recorded in the embedded detector card. The input index and exact source paths are specified in `docs/sipm_electronics_handoff.md`. Each source has its own content identity, event identifiers and provenance.

The upstream calculation includes electron diffusion, attachment, electroluminescence production and geometrical photon collection. Its adopted EL channel is 5 mm in diameter and depth, at 15 bar and $E/P = 3\ \text{kV}\,\text{cm}^{-1}\,\text{bar}^{-1}$. The adopted uniform emission along the electron path motivates a uniform conditional photon arrival time within each bucket. This is a timing approximation at the response interface. It cannot restore sub-bucket timing that was not saved by the producer. The response preserves the producer's signed bucket index and geometrical reference.

A *primary photodetection* is the successful conversion of an incident photon into an avalanche trigger. A *dark pulse* is an avalanche initiated without an incident signal photon. The integer number of triggers differs from charge expressed in *photoelectron equivalents* (PE): one PE denotes the calibrated charge q_0 of a fully recovered avalanche. A partially recovered avalanche contributes a fraction of one PE. We use *microcell* for an avalanche element inside a SiPM and *channel* for a complete SiPM.

2 Sensor reference and adopted operating point

Hamamatsu's AXEL presentation describes a custom VUV-sensitive device based on the S13370 family, with 50 μm microcell pitch, TSV connections and a 64-channel assembly on a flexible polyimide substrate with a PEEK stiffener [1]. The displayed PDE curve was taken at $-25\ ^\circ\text{C}$ and 4 V overvoltage; it is not a calibration of our room-temperature design. The AXEL prototype analysis names the standard S13370-3050CN [2]. That standard device has a $3 \times 3\ \text{mm}^2$ active area and 3600 microcells.

Our 5 mm diameter active disk follows the optical geometry already used in photon production. Its effective microcell count is a geometrical design approximation.

Parameter	Adopted input	Status or origin
Bucket width Δt	250 ns	User design input; detector card
Photon detection efficiency p	0.30	User design input
Active disk diameter d	5 mm	Existing optical geometry
Microcell pitch a	50 μm	AXEL device family reference
Microcell count M	round$[\pi(d/2)^2/a^2]$	Effective uniform-cell geometry
Fully recovered gain G	2.55×10^6	Standard-device reference
Recovery time τ	73.4 ns	User adopts AXEL measured mean
Reference dark rate R_0	10^6 s^{-1} per 9 mm^2	Typical reference at 25 °C, 4 V
ASIC noise σ_{PE}	0.5 PE RMS/bucket	User provisional input
ASIC upper range $I_{\max}$	1 mA	User provisional input

Table 1: Baseline inputs. Runtime choices reside in versioned TOML cards. Gain and dark-rate references are provisional standard-device values; they do not characterize a fabricated 5 mm circular device.

The gain and dark-rate reference comes from the search-indexed manufacturer specification KSX-I50133-E [3]. The original PDF remains to be recovered; this source limitation is recorded in the bibliography and specification note. The table gives a typical dark rate of 1 MHz and a maximum of 3 MHz for the standard device at 25 °C and 4 V overvoltage. The baseline takes the typical value as an independent primary-dark-avalanche rate. Its measurement definition must be checked before introducing separately sampled correlated avalanches. The same source lists 3% crosstalk; the baseline probability is nevertheless set to zero as an explicit modeling approximation. A suitable room-temperature afterpulse probability and delay distribution has not been established.

The PDE includes optical fill factor and avalanche initiation probability in its calibrated definition. The reference dark rate also already reflects the reference device's geometry. Neither receives an additional fill-factor multiplier. Temperature and overvoltage are recorded as reference operating conditions; a transfer law for these parameters is not assumed.

2.1 Parameters exported from the saved calculation

The following values are generated from the pilot HDF5 output. In particular, the dark-rate scaling, one-PE charge, noise-current conversion and equivalent current range are evaluated by the Julia calculation using the selected cards.

Parameter	Value	Unit
PDE	0.3	
Effective microcells	7854	
Gain	2550000	electrons/PE
Recovery time	7.34e-08	s
Sensor diameter	0.005	m
Primary dark rate	2181662	Hz
Charge per fully recovered PE	4.08555e-13	C
Bucket duration	2.5e-07	s
ASIC noise RMS	0.5	PE
ASIC noise RMS	8.171101e-07	A
Upper average-current limit	0.001	A
Equivalent upper range	611.9127	PE/bucket
Mean primary dark pulses	0.5454154	per bucket

3 Stochastic microcell response

3.1 Detection, arrival times and illumination

At fixed incident count, sample the number of primary photodetections as

$$N_{s,cb} \mid N_{\gamma,cb} \sim \text{Binomial}(N_{\gamma,cb}, p). \tag{1}$$

Conditional on $N_{s,cb}$, draw each time independently and uniformly over $[t_b, t_b + \Delta t)$, and assign each detection independently to one of the M microcells with probability $1/M$. This is equivalent to sampling times and microcell labels for every incident photon and retaining independent Bernoulli detections, while avoiding work for undetected photons. Uniform illumination is an additional sensor approximation: the saved input resolves SiPM channels, not the illumination pattern inside one sensor.

The baseline holds p fixed during microcell recovery. Consequently a successful trigger at short delay still initiates an avalanche, whose charge is reduced by the law below. A model in which the initiation probability also depends on recovered overvoltage would need an additional calibrated function $p(r)$; the AXEL mean correction alone does not determine that function.

3.2 Recovery and avalanche charge

Let $t_{\text{last},j}$ be the time of the previous avalanche in microcell j. Just before a trigger at t_k, its recovered fraction and charge are

$$r_j(t_k^-) = 1 - \exp[-(t_k - t_{\text{last},j})/\tau], \tag{2}$$

$$q_k = q_0 r_j(t_k^-), \qquad q_0 = eG. \tag{3}$$

After the avalanche set $t_{\text{last},j} = t_k$. An explicitly fully recovered microcell has $t_{\text{last},j} = -\infty$. The default dark-equilibrium initial state is specified below. The state is carried continuously between buckets; a bucket boundary does not recharge the sensor. Equation (3) follows the exponential gain-recovery model used by AXEL [4]. Charge accumulation and exponentials use SI units and Float64 arithmetic in the implementation.

The time constant is not a fixed dead interval. At a delay τ, the recovered charge is $(1 - e^{-1})q_0$; after 2τ and 3τ the fractions are $1 - e^{-2}$ and $1 - e^{-3}$. Full recovery is approached asymptotically. The probability to initiate a second avalanche and its available charge are distinct quantities; this baseline specifies the former by a constant PDE and the latter by equation (3).

For all avalanches assigned to channel c and bucket b,

$$Q_{s,cb} = \sum_{k \in \text{signal}} q_k, \qquad Q_{d,cb} = \sum_{k \in \text{dark}} q_k, \tag{4}$$

$$Q_{cb} = Q_{s,cb} + Q_{d,cb}, \qquad m_{cb} = Q_{cb}/q_0, \qquad \overline{I}_{cb} = Q_{cb}/\Delta t. \tag{5}$$

The signal and dark sums label the cause of each avalanche. Because both populations change the same microcell state, Q_s is affected by preceding dark avalanches and Q_d by preceding signal avalanches. These saved components provide charge accounting, rather than independent responses that could be added after simulating two isolated sensors.

3.3 Where the fluctuations enter

For known microcell labels and arrival times, equation (3) is deterministic. In an ensemble, the labels, times, primary photodetections and dark counts vary. Two detected photons may strike different cells and both produce full charge, or strike the same cell and produce $q_0 + q_0[1 - \exp(-\delta t/\tau)]$ if it was

initially recovered. Their charge therefore fluctuates even at fixed detected-photon count. Additional intrinsic single-avalanche gain dispersion is set to zero in the baseline.

At low occupancy and in the absence of dark pulses, the recovered-cell limit provides a useful check:

$$\mathbb{E}[Q \mid N_\gamma] = q_0 p N_\gamma, \qquad \mathrm{Var}(Q \mid N_\gamma) = q_0^2 p(1-p) N_\gamma. \tag{6}$$

This is the variance conditional on the supplied photon count. Upstream photon-production and collection fluctuations are already represented by the input event ensemble. They must not be resampled as an extra Poisson input.

4 Dark pulses and the sensor's dark current

The adopted area scaling at fixed temperature, overvoltage and technology is

$$A = \pi(d/2)^2, \qquad R_d = R_0 \frac{A}{A_0}, \qquad A_0 = 9 \text{ mm}^2. \tag{7}$$

For each selected channel and acquisition bucket draw $N_{d,cb} \sim \mathrm{Poisson}(R_d \Delta t)$, with uniform times and uniform microcell labels. Merge these triggers with photodetections in time order before updating the cell states. Equivalently one can generate a homogeneous Poisson process over the acquisition interval. The time-window and initial-state choices are part of the response card and provenance. Dark pulses occur in buckets with zero incident photons as well as signal buckets.

The default acquisition uses every channel illuminated anywhere in the event. All selected channels are simulated over the same interval: the global first to last occupied photon bucket, with four additional buckets on each side (1 μs at the default sampling). This defines the channel set for the pilot; it does not include the unilluminated remainder of the detector. The sensor card can add explicitly listed channels. For an empty photon history it defines the first and last acquisition bucket, supporting a dark-only acquisition. No threshold is applied to the output samples.

At acquisition start t_0, initialize each microcell with an independent age $U_j \sim \mathrm{Exponential}(\mathrm{mean} = M/R_d)$ and set $t_{\mathrm{last},j} = t_0 - U_j$. This is the stationary backward-recurrence age for the assumed Poisson dark triggers. It represents the dark activity before the recorded window without storing those earlier pulses. At zero dark rate the cells start fully recovered. This initialization assumes the signal begins within the recorded photon interval, with only dark activity preceding it.

In the dilute, recovered-cell limit, a dark-only exposure of duration T has

$$\mathbb{E}[Q_d] = q_0 R_d T, \qquad \mathrm{Var}(Q_d) = q_0^2 R_d T, \qquad \mathbb{E}[I_d] = q_0 R_d. \tag{8}$$

Thus dark current is the time-averaged charge carried by random dark pulses, and the pulse fluctuations survive subtraction of its expectation. At finite occupancy recovery reduces both pulse amplitudes and the average dark current; the stationary result is derived below. A continuous leakage current would be a separate baseline term, currently taken as zero. Independent dark-count processes across channels and times are an approximation to the primary thermal avalanche population. Optical crosstalk and afterpulsing would add correlated triggers and require separate calibrated parameters.

5 Current readout and the provisional ASIC

The SiPM supplies avalanche current pulses; a voltage signal arises only through the attached electrical network. For the current bucket model, each avalanche contributes its integrated charge to its bucket, so the observable is $\overline{I} = Q/\Delta t$. This convention preserves

$$\sum_b \overline{I}_{cb} \Delta t = \sum_k q_k \tag{9}$$

for avalanches inside the acquisition interval. The model specifies neither an instantaneous peak current nor a shaped voltage waveform. A finite pulse shape and front-end transfer function would redistribute charge across buckets and require a revised readout calculation with explicit boundary losses.

With known zero baseline, the adopted ASIC model is

$$\epsilon_{cb} \sim \mathcal{N}(0, \sigma_{\mathrm{PE}}^2), \tag{10}$$

$$I_{cb}^{\mathrm{raw}} = \overline{I}_{cb} + \epsilon_{cb} \frac{q_0}{\Delta t}, \tag{11}$$

$$I_{cb}^{\mathrm{out}} = \min(I_{cb}^{\mathrm{raw}}, I_{\max}), \tag{12}$$

$$s_{cb} = \mathbf{1}\{I_{cb}^{\mathrm{raw}} > I_{\max}\}. \tag{13}$$

Noise is independent between channels and buckets. It is added before the upper range limit. Negative baseline-subtracted samples are retained; the model has no lower clipping threshold. Save the sampled noise, current before clipping, output current and saturation flag. The independent sensor output remains available for changing or diagnosing the electronics model.

The conversion and range in PE units follow from the selected gain:

$$I_{1\mathrm{PE}} = \frac{q_0}{\Delta t}, \quad \sigma_I = \sigma_{\mathrm{PE}} \frac{q_0}{\Delta t}, \quad m_{\max}^{\mathrm{ASIC}} = \frac{I_{\max} \Delta t}{q_0}. \tag{14}$$

The fixed hardware input is 1 mA. Its PE equivalent changes with G and Δt. This specification supersedes the earlier approximate 300-PE range discussed before the ASIC current limit was adopted. It always applies to the sensor's measured charge/current response, after photon detection, with the Gaussian term added as in equation (13).

5.1 Relation to AXEL readout and noise measurements

AXEL's front-end electronics use a 5 MS/s physics readout, with a 17 ns first-order stage followed by an approximately 400 ns second-order Sallen–Key stage. A 40 MS/s auxiliary readout supports single-PE dark-pulse gain calibration [5]. Its physical sampling interval is 200 ns, whereas the adopted NAUSICAA interval is 250 ns. These transfer functions do not define our ASIC.

The 2025 AXEL paper reports a mean baseline standard deviation of 0.46 ADC counts during operation and 0.45 without high voltage, with a 12-bit ADC and 0.488 mV per ADC count [2]. The 2024 analysis uses a 3.5-ADC threshold, corresponding to 3.8 detected-photon units, and extends integration by five samples on each side of the threshold crossings. It also considers a baseline-offset uncertainty within one ADC count separately [6]. A persistent baseline offset and independent sample noise propagate differently when integrating an event.

Combining a PE calibration from one run with baseline RMS from another only supplies a rough comparison. Our 0.5-PE Gaussian term is the adopted ASIC input, additional to explicitly generated sensor dark pulses. It is not a measured separation of AXEL's baseline into dark, analog and digitization components. Shaping-induced correlations, offset drift and ADC quantization remain extensions requiring ASIC information.

6 Mean linearity and subsequent signal recovery

6.1 AXEL's stationary mean response

The AXEL correction can be derived from the same exponential charge law under steady uniform illumination [4]. Let the combined successful signal and dark triggers in each microcell form a stationary

Poisson process of rate λ. Its interval density is $f(u) = \lambda e^{-\lambda u}$ for $u \geq 0$. Averaging the recovered fraction at avalanche times gives

$$\mathbb{E}[r] = \int_0^\infty (1 - e^{-u/\tau})\lambda e^{-\lambda u}\, du = \frac{1}{1+\lambda\tau}. \tag{15}$$

For expected primary avalanche count $n = M\lambda\Delta t$, the mean charge in PE units is

$$m = \frac{n}{1+kn}, \qquad k = \frac{\tau}{M\Delta t}. \tag{16}$$

The low-intensity limit is $m \simeq n - kn^2$. The large-rate stationary current approaches Mq_0/τ. This stationary asymptote is not a bound on the first short bucket of an initially charged sensor, which also contains stored charge. Sensor nonlinearity depends on occupancy per microcell and recovery time; electronics clipping depends on the configured current limit.

Dark-only stationary current follows by setting $\lambda = R_d/M$:

$$\mathbb{E}[I_d] = \frac{q_0 R_d}{1 + R_d\tau/M}. \tag{17}$$

Equations (15)–(17) assume constant rate, independent triggering and fixed initiation probability. The simulation conditions on the supplied finite counts and follows state across time-varying buckets, so the stationary expression is a validation limit and a candidate reconstruction law, not an identity for every event.

6.2 Compact recovery-state and charge moments

The following local derivation establishes the basis for a faster stochastic bucket response. It uses the implemented assumptions: uniform independent microcell selection, fixed initiation probability, exponential recovery, and reset to zero recovery after every successful trigger. The current production implementation follows individual avalanches; the compact model below is a proposed implementation strategy.

Exact closure of the mean state. Let $r_j \in [0,1]$ be the recovered fraction of microcell j and define $R = \sum_{j=1}^{M} r_j$. Between avalanches,

$$\frac{dR}{dt} = \frac{M - R}{\tau}. \tag{18}$$

At an avalanche, a uniformly selected cell J contributes r_J PE and resets to zero. Conditional on the entire state $\boldsymbol{r}$,

$$\mathbb{E}[\Delta m \mid \boldsymbol{r}] = \frac{R}{M}, \tag{19}$$

$$\mathbb{E}[R^+ \mid \boldsymbol{r}] = \left(1 - \frac{1}{M}\right) R^-. \tag{20}$$

Consequently, conditional on a sequence of arrival times, the ensemble mean of R and the expected collected charge evolve using only the mean initial R. This closure is exact for these assumptions. Averaging over arrival times preserves it. A distribution of recovery states is unnecessary for calculating the mean charge.

The specific stochastic contribution. Random microcell selection contributes the conditional avalanche-charge variance

$$\mathrm{Var}(\Delta m \mid \boldsymbol{r}) = \frac{1}{M}\sum_{j=1}^{M} r_j^2 - \left(\frac{R}{M}\right)^2. \tag{21}$$

A deterministic charge assignment $\Delta m = R/M$ removes this contribution. An aggregate stochastic response can retain it through charge moments. The same avalanche changes the outgoing recovery state, which creates a correlation between collected charge and the response to subsequent light.

Closed moment equations in the Poisson limit. For independent Poisson triggers of deterministic rate $\lambda(t)$ per microcell, let $u_1 = \mathbb{E}[r]$ and $u_2 = \mathbb{E}[r^2]$. The recovery drift and reset give

$$\dot{u}_1 = \frac{1-u_1}{\tau} - \lambda u_1, \tag{22}$$

$$\dot{u}_2 = \frac{2(u_1-u_2)}{\tau} - \lambda u_2. \tag{23}$$

At constant rate, their stationary values are

$$u_1 = \frac{1}{1+\lambda\tau}, \qquad u_2 = \frac{2}{(1+\lambda\tau)(2+\lambda\tau)}. \tag{24}$$

The first reproduces equation (15).

Let q denote accumulated charge in PE units from one microcell during the current bucket, and write $a = \mathbb{E}[q]$, $b = \mathbb{E}[qr]$ and $c = \mathbb{E}[q^2]$. At a trigger $(q, r) \mapsto (q + r, 0)$, yielding

$$\dot{a} = \lambda u_1, \tag{25}$$

$$\dot{b} = \frac{a-b}{\tau} - \lambda b, \tag{26}$$

$$\dot{c} = \lambda(2b + u_2). \tag{27}$$

Start each bucket with $a = b = c = 0$ and the incoming recovery moments. At its end, the single-cell charge variance is $c - a^2$ and the covariance of collected charge with final recovery is $b - au_1$. For independent microcells, the total mean and variance are Ma and $M(c - a^2)$, and the covariance with total outgoing recovery is $M(b - au_1)$. Thus a finite set of moment equations determines these observables in the Poisson model, including the effect of collection throughout the bucket.

Production interpretation. The implemented optical counts are binomial conditional on saved photons; arrival times are uniform conditional on those counts. A Poisson replacement would change the prescribed count fluctuations. The compact production model should preserve those count draws and derive or calibrate its conditional charge and outgoing-state moments accordingly. The equations above provide an exact analytical limit and a practical starting point for that calculation. Its stochastic sampler must reproduce the charge variance and its covariance with outgoing recovery. Validation observables are bucket means and variances, neighbouring-bucket covariance, and the total-event charge distribution. A detailed recovery-state distribution should be introduced only if these observables demonstrate a consequential failure of the compact model.

6.3 Conditional-count statistical implementation

The prototype in `src/ion_signal/sipm_statistical.jl` implements a count-preserving discretization of the moment model. It retains the binomial photodetection and Poisson dark-count draws. The state carried between buckets is (R, S), with $S = \sum_j r_j^2$. Dark-equilibrium initialization samples the cell recoveries once and reduces them to these two totals.

Exact conditional occupancy moments. For k simultaneous triggers assigned independently to M cells, write

$$a = (1 - 1/M)^k, \qquad b = (1 - 2/M)^k, \qquad p = 1 - a. \tag{28}$$

Each occupied cell contributes its available charge once. For interval charge Q in PE equivalents,

$$\mathbb{E}[Q \mid R, S, k] = pR, \tag{29}$$

$$\mathrm{Var}(Q \mid R, S, k) = (a - b)S + (b - a^2)R^2, \tag{30}$$

$$\mathbb{E}\left[\sum_{j \text{ hit}} r_j^2 \mid R, S, k\right] = pS. \tag{31}$$

These expressions follow from the single-cell occupancy probability p and the two-cell occupancy covariance $b - a^2$. They condition on the actual trigger count. The implementation uses a numerically stable equivalent of the variance expression and handles zero and one trigger explicitly.

Charge and outgoing state. The interval charge uses a bounded beta draw matching the first two moments in equation (31). Its support respects the total available charge, trigger count and the maximum cell recovery allowed by (R, S). Zero variance gives a deterministic draw; maximal bounded variance uses a two-point distribution. The outgoing first moment is $R^+ = R^- - Q$, so output charge and recovery loss are correlated by construction. For a single trigger the second-moment loss is exactly Q^2. For a batch, the closure uses $W = QS/R$ for $R > 0$, giving $\mathbb{E}[W] = pS$ and $S^+ = S^- - W$. This preserves the conditional mean loss of S. Its higher joint moments are an approximation. Feasibility repairs enforcing $R^2/M \le S \le R$ and bounded-variance limiting are recorded as numerical diagnostics.

For fixed interval counts, the first two moments of cumulative charge close in $\mathbb{E}[R]$, $\mathbb{E}[S]$, $\mathbb{E}[R^2]$, $\mathbb{E}[Q_{\text{tot}}]$, $\mathbb{E}[Q_{\text{tot}}R]$ and $\mathbb{E}[Q_{\text{tot}}^2]$. Indeed, the charge mean is linear in R, its second moment is linear in S and R^2, and recovery is affine in the individual cell recoveries. Preserving the conditional charge moments and $\mathbb{E}[W] = pS$ therefore preserves those aggregate moments for the discretized process, provided no feasibility repair or variance limiting modifies the update. The beta distribution approximates the shape and higher moments of that process; it does not introduce an additional approximation to these closed first and second moments.

For mixed optical and dark triggers, labels are exchangeable. Conditional on the interval charge and squared-charge sum, random assignment of k_s signal labels among $k = k_s + k_d$ triggers gives

$$\mathbb{E}[Q_s \mid Q, W] = \alpha Q, \qquad \alpha = k_s/k, \tag{32}$$

$$\mathrm{Var}(Q_s \mid Q, W) = \frac{\alpha(1 - \alpha)}{k - 1}(kW - Q^2), \quad k > 1. \tag{33}$$

The prototype substitutes the closed estimate of W, constrained to its conditional bounds for this label allocation, and samples bounded Q_s; $Q_d = Q - Q_s$. This preserves the total sampled charge and both population charge bounds. For k trigger amplitudes in $[0, 1]$ summing to Q, these conditional bounds are $Q^2/k \le W \le \lfloor Q \rfloor + (Q - \lfloor Q \rfloor)^2$. Optical/dark partition moments inherit this approximation to W; the saved production signal uses their sum Q.

Arrival-time discretization. Dense buckets are split into K equal intervals. Successive binomial partitions give the exact multinomial allocation of the fixed optical and dark counts to these intervals. Half an interval of exponential recovery precedes the occupancy draw and half follows it. Collapsing arrivals within an interval to its midpoint is the time-discretization approximation; comparison of $K = 32, 64, 128$ quantifies it. Buckets with at most 16 triggers instead sample their individual uniform arrival times and apply successive single-trigger moment draws. They retain the same compact state.

The TOML selector `response_model="statistical"` chooses this path; `statistical_substeps` and `sparse_trigger_limit` specify its resolution. The prototype card uses 128 intervals per 250 ns bucket and a sparse limit of 16 triggers. The microscopic path remains the reference. Model cards and code hashes distinguish the saved responses. The validation script `scripts/production/benchmark_sipm_statistical.jl` compares transient means, variances, adjacent-bucket covariance, total event charge and runtime for synthetic histories and the saved photons from both operating points.

6.4 Inverse correction and what it recovers

Solving equation (16) for n gives AXEL's correction

$$\widehat{n} = F(m) = \frac{m}{1 - km}, \qquad km < 1. \tag{34}$$

AXEL measured device recovery with an LED and a low-intensity reference, then used individual recovery times in the nonlinearity correction. The 2024 study reports a mean of 73.4 ns for its devices, with 3600 microcells and 200 ns sampling [6]. It also adjusts the effective recovery calibration using photopeaks and a corrected squared-sum observable; the 2025 analysis develops this fine-tuning alongside EL-gain, time and drift-distance corrections [2]. Our initial forward-model τ is the adopted mean; a later reconstruction calibration may fit an effective value to the simulated pulse population.

The correction maps measured charge to an estimate of the primary avalanche count under its rate assumptions. It cannot identify which photons were missed, recover the random microcell collision history, or remove its event fluctuations. Even with exact mean calibration, $\mathbb{E}[F(m_{\text{event}})]$ need not equal $F(\mathbb{E}[m_{\text{event}}])$. For small fluctuations around m_0, its local noise amplification is

$$F'(m_0) = \frac{1}{(1 - km_0)^2}, \qquad \operatorname{Var}(\widehat{n}) \simeq \frac{\operatorname{Var}(m)}{(1 - km_0)^4}. \tag{35}$$

The positive curvature produces a leading mean shift $k \operatorname{Var}(m)/(1 - km_0)^3$ in this small-fluctuation approximation. Both expressions fail near the pole, where the full response distribution must be used. Correcting a clipped output cannot determine the lost current. Those samples retain their saturation flag and require an explicitly justified censored-data or event-selection treatment in a later analysis.

6.5 Dark subtraction, PDE and energy analysis

With stationary signal plus dark counts, equation (34) estimates their *total* primary count. In that limit a possible signal estimator is

$$\widehat{n}_s = F(m) - R_d \Delta t. \tag{36}$$

Subtracting a full-charge mean dark signal before the nonlinear inversion is generally inequivalent, because signal and dark avalanches compete for the same recovering cells. For a time-varying event, dark-only calibration and joint forward simulations should establish the subtraction and inversion bias. Subtracting a known mean dark contribution leaves its statistical variance. The saved signal/dark charge components permit direct checks of these effects in simulation, while a real acquisition measures their sum.

After validating the correction, dividing $\widehat{n}_s$ by the calibrated PDE estimates incident photons. The division changes the scale and preserves the detection fluctuation. Converting incident light to EL electrons and deposited energy adds the separate optical collection, EL gain and attachment calibrations already associated with the upstream gas model. These stages should be tested against saved truth quantities in order: sensor charge, primary detected count, incident photons and event energy. The inverse model is a subsequent analysis stage; it does not alter the saved forward sensor or ASIC histories.

7 Implementation, persisted stages and validation

The photon reader is `read_photon_history` in `src/event_analysis/production_readout.jl`, exported through `EventAnalysis.jl`. It returns owned arrays with signed SiPM and bucket indices. The `Sensor` type in `src/ion_signal/sipm_response.jl` stores PDE, microcell count, gain, recovery time and dark rate. Its `simulate_channel` function samples successive photon buckets and returns owned count and charge arrays. The `Electronics` type in `src/ion_signal/sipm_electronics.jl` stores sampling interval, noise and range; `apply_electronics` returns owned noise/current/flag arrays while preserving its input charge. The diagnostic `axel_mean` and `axel_inverse` functions evaluate the stationary formulas; the inverse reports a validity flag and rejects clipped samples or a nonpositive denominator.

The sensor and electronics cards are independently versioned at `config/specs/response/sipm/sipm_v1.toml` and `config/specs/response/sipm_electronics/readout_v1.toml`. The acquisition adapter and persisted stages are implemented in `scripts/production/sipm_pipeline.jl`, with `scripts/production/run_sipm_pilot.jl` as the Julia pilot entry point. A production run card selects the inputs and stages. Canonical card hashing and event-keyed seeds follow `scripts/production/specs.jl`. The sensor file schema is `nausicaa.sipm_file.v1`, with event schema `nausicaa.sipm_history.v1`; the ASIC file uses `nausicaa.electron_response_file.v1`, with event schema `nausicaa.electron_response_history.v1`. Metadata identify the model, algorithm, units and initialization convention explicitly.

For each saved channel–bucket cell, the sensor stage records incident photons, primary signal and dark avalanche counts, signal and dark charge, and total charge. The saved q_0 and Δt determine PE-equivalent response and bucket-average current exactly from the charge arrays. The output must also describe the acquisition channel set and window so that absent photon input cells can be represented with dark and electronics noise. The electronics stage references the saved sensor identity and stores its noise draw, current before and after clipping, and saturation flag. This separation permits ASIC changes while preserving the identical sampled sensor realization.

Source-content hashes, event and gas identifiers, parent-pair checksum, geometry/time conventions, exact TOML text and canonical hashes, RNG seeds, and code identities accompany the outputs. Output files are marked complete only after successful writing. Physical calculations use Julia and save results to HDF5; Python presentation code reads those results to generate parameter tables and this report's numerical additions. The four source files are preserved, and the response is first evaluated on a small pilot.

The validation program tests physical limits and persisted behavior:

1. Check the binomial mean and variance at negligible occupancy, and one- and two-avalanche charge against equation (3).

2. Check recovery through empty buckets and a signal at a bucket boundary; changing bucket labels must not reset the microcells.

3. Check Poisson dark-count mean, variance and area scaling, with charge approaching equation (8) in the dilute limit.

4. Compare a constant-rate simulation after equilibration with the stationary mean in equation (16); then quantify deviations for finite pulses and changing illumination.

5. Check equation (9) and verify that saved signal and dark components sum to the sensor response.

6. Check the Gaussian noise distribution, independence prescription, upper clipping and retained negative samples, including zero illumination.

7. Round-trip both HDF5 stages, verify identities and event seeds, and confirm that changing only ASIC parameters preserves the sensor realization.

8. On the four-input pilot, report channel/bucket occupancy, charge loss from recovery, dark contribution, clipped fraction and resource use before scaling to all events. Later inversion tests report bias and resolution as functions of intensity, pulse shape and distance from clipping.

The active approximations are uniform sub-bucket timing and illumination, fixed PDE during recovery, common microcell gain and recovery time, independent primary dark pulses, zero correlated avalanches, and a charge bucket ASIC with independent Gaussian noise. They define a reproducible baseline. Sensor measurements and ASIC information can replace each input through the versioned cards while keeping the saved stage boundaries and truth comparisons explicit.

8 Saved pilot and validation results

The pilot uses event 0 in each of the four existing 250 ns photon histories. Channels illuminated anywhere in the event are acquired through the global photon-bucket span with four quiet buckets at either end. Zero-photon buckets contain dark pulses and ASIC noise. Recovery loss includes optical and dark charge. Clipping counts include every acquired channel–bucket cell. These four events establish operation of the pipeline; they do not estimate the full-sample saturation fraction.

Sample	Gas	Event	Detected PE	Recovery loss	Clipped buckets
bb0nu	pure_xe	0	13,782,681	0.12357	5,038 / 44,652
bb0nu	xe_he15	0	13,558,437	0.28869	2,360 / 18,550
single_electrons	pure_xe	0	14,493,517	0.23641	3,151 / 18,850
single_electrons	xe_he15	0	14,255,194	0.39132	1,552 / 9,088

bb0nu, pure_xe: maximum sensor charge 11697.641 PE; dark pulses 24,713; ASIC noise RMS 0.49769 PE.

bb0nu, xe_he15: maximum sensor charge 19665.466 PE; dark pulses 10,135; ASIC noise RMS 0.50167 PE.

single_electrons, pure_xe: maximum sensor charge 16097.155 PE; dark pulses 10,425; ASIC noise RMS 0.50524 PE.

single_electrons, xe_he15: maximum sensor charge 20557.517 PE; dark pulses 5,039; ASIC noise RMS 0.49691 PE.

8.1 Independent numerical checks

The two-trigger expectation is independent midpoint quadrature of the uniform-arrival separation density. The stationary test uses independent Poisson photon counts, 40 microcells and 80 detected triggers per bucket; the transient test begins with recovered cells. The inverse means demonstrate finite-bin bias and the difference between stationary and transient illumination. Dark-count tests use the 5 mm effective sensor; binomial tests use 100 incident photons at PDE 0.30. ASIC tests include variance, lag-one covariance, charge/current units and exact range boundaries. Interface checks include signed gaps, empty events, HDF5 round trips, seed replay and reuse of the sensor output.

Validation quantity	Saved value
axel_mean_pe	50.40323
binomial_mean	30.0355
binomial_variance	21.1201
dark_mean	0.54458
dark_variance	0.5446029
electronics_noise_mean_pe	-0.0009798307
electronics_noise_rms_pe	0.5003612
float32_charge_storage_relative_error	8.516565e-11
stationary_inverse_mean_pe	80.57573
stationary_inverse_rms_pe	11.31597
stationary_mean_pe	50.39924
transient_inverse_mean_pe	95.3881
transient_mean_pe	56.05311
two_trigger_mean	1.577185
two_trigger_quadrature	1.579483
two_trigger_variance	0.07313562

The pilot demonstrates substantial clipping under the adopted 1 mA bucket-average-current limit. The saved sensor stage retains the charge that precedes this loss, allowing the current range and reconstruction strategy to be reviewed independently. A recovery correction alone cannot restore the clipped samples. The stationary and transient tests also show why the inverse requires calibration for the time structure of the actual light pulses. The completed 8,000-event bb0nu response and the operating-point comparison are documented in `docs/response_overview.md`.

The statistical implementation uses the same production PE-per-bucket schemas: `nausicaa.sipm_pe_file.v1` for sensor charge and `nausicaa.sipm_readout_pe_file.v1` for ASIC response. Its card is `config/specs/response/sipm/sipm_statistical_v1.toml`. The source, interfaces, validation commands and two-event integration pilot are described in `docs/sipm_statistical_implementation.md`. The generated comparison is `latex_generated/sipm_statistical_validation/validation.pdf`, with an HTML counterpart at `results/sipm_statistical_validation/validation.html`.

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